DSA06 DATA HANDLING AND VISUALIZATION

Activity: Identify Seasonality Patterns | Slot D | Series SP-D12: Clinic visits

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Question Paper

DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-D12: Clinic visits [20 marks]

Scenario: You are the analyst for a community health clinic. The series visits (Patient visits) is recorded monthly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 12 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(67699)
m <- 12                      # observations per cycle (monthly data)
n <- m * 5                  # 5 full cycles
t <- 1:n
trend <- 378.8 + 2.332 * t
visits <- round(trend + 88.4 * cos(2 * pi * (t - 9) / m) + rnorm(n, 0, 19.9), 0)
x <- ts(visits, frequency = m, start = c(2021, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(visits ~ 1:n) and report the slope per month). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 12 seasonal values). Name the peak month and the trough month with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 5 cycles is the highest value in the peak month from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(visits ~ 1:n))) and compute acf(d, lag.max = 12, plot = FALSE) using observation lags. Report the autocorrelation at lag 12 and at lag 6, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of visits in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per month [2]. Write one sentence of advice to management based on your findings [2]. [6]

Series Creation (given code, not edited)

```
set.seed(67699)
m <- 12
n <- m * 5
t <- 1:n
trend <- 378.8 + 2.332 * t
visits <- round(trend + 88.4 * cos(2 * pi * (t - 9) / m) + rnorm(n, 0, 19.9), 0)
x <- ts(visits, frequency = m, start = c(2021, 1))
```

Output: `print(x)`

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2021	351	290	300	308	340	402	468	438	482	466	462	414
2022	364	336	329	344	360	439	459	506	520	497	511	405
2023	429	390	378	353	419	471	514	524	529	525	481	475
2024	400	400	366	409	438	453	522	589	558	517	531	526
2025	433	393	401	420	437	536	562	584	608	571	569	511

Task 1. Look at the series [3 marks]

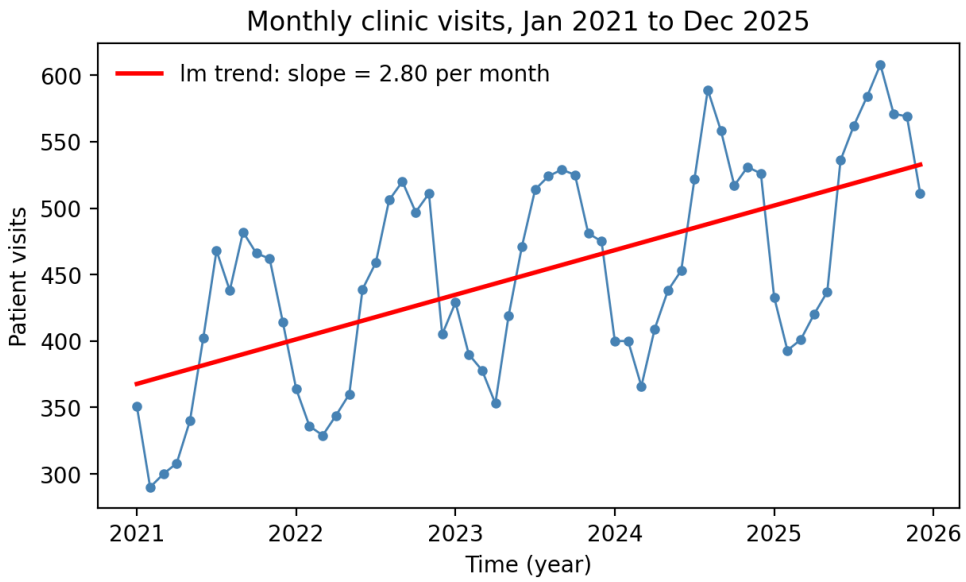
R code

```
length(x)
frequency(x)
length(x) / frequency(x)
fit <- lm(visits ~ 1:n)
coef(fit)[2]

plot(x, type = "o", pch = 16, col = "steelblue",
     main = "Monthly clinic visits, Jan 2021 to Dec 2025",
     xlab = "Time (year)", ylab = "Patient visits")
lines(time(x), fitted(fit), col = "red", lwd = 2)
```

Output

`[1] 60` (n) `[1] 12` (cycle length) `[1] 5` (full cycles) `1:n 2.80` (slope)



Answer:

n = 60 observations, cycle length = 12 (monthly data), so there are 60 / 12 = 5 full cycles.

The slope of lm(visits ~ 1:n) is 2.80 visits per month. It is positive, so the overall trend is rising.

Task 2. The seasonal pattern [4 marks]

R code

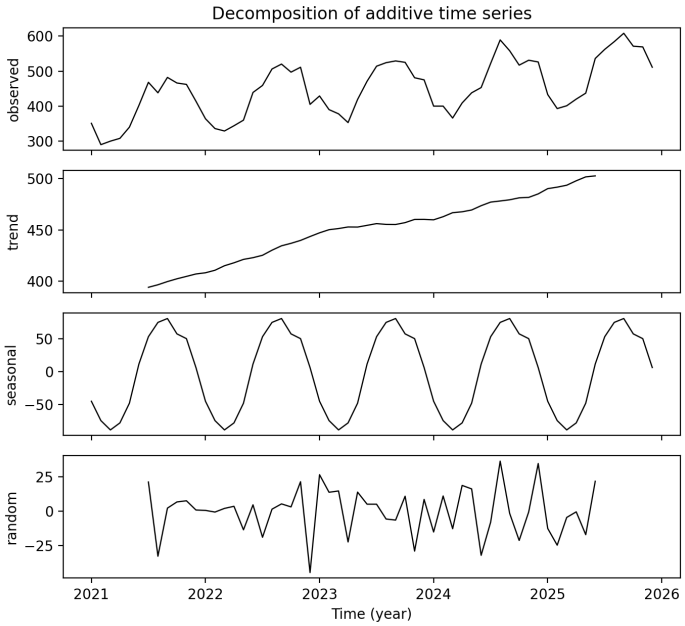
```
dec <- decompose(x)
fig <- dec$figure
names(fig) <- month.abb
round(fig, 2)
names(which.max(fig)); max(fig)
names(which.min(fig)); min(fig)
max(fig) - min(fig)

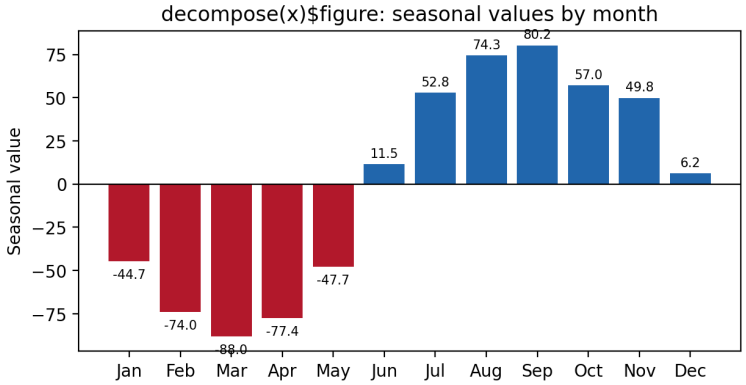
plot(dec)
barplot(fig, col = ifelse(fig >= 0, "steelblue", "firebrick"),
        main = "Seasonal values by month")
```

Output: `round(fig, 2)`

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
-44.71	-73.96	-88.04	-77.45	-47.65	11.47	52.77	74.35	80.22	57.00	49.82	6.17

Peak: `Sep 80.22` Trough: `Mar -88.04` Swing: `168.26`





Answer:

Peak month = September, seasonal value = +80.22.

Trough month = March, seasonal value = -88.04.

Seasonal swing = 80.22 - (-88.04) = 168.26 visits.

Task 3. Is it consistent? [3 marks]

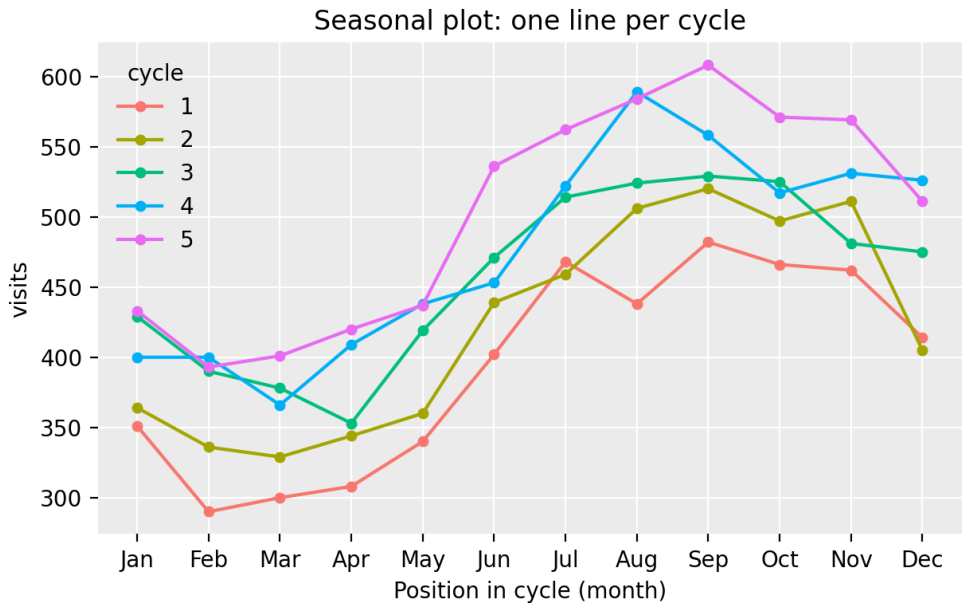
R code

```
library(ggplot2)
df <- data.frame(visits = as.numeric(x),
                 pos = as.numeric(cycle(x)),
                 cycle = factor(rep(1:5, each = 12)))
ggplot(df, aes(pos, visits, group = cycle, colour = cycle)) +
  geom_line() + geom_point() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(title = "Seasonal plot: one line per cycle",
       x = "Position in cycle (month)", y = "visits")

peak <- which.max(dec$figure)
tapply(df$visits, df$cycle, which.max)
sum(tapply(df$visits, df$cycle, which.max) == peak)
```

Output

Month of the highest value in each cycle: `1 = 9, 2 = 9, 3 = 9, 4 = 8, 5 = 9`
`[1] 4` (number of cycles with the highest value in the peak month)



Answer:
The highest value is in the peak month (September) in 4 out of the 5 cycles. In cycle 4 the highest value is in August (589), but September (558) is also high there.
So yes, the pattern is consistent. In every cycle the visits are low in Feb to Apr, go up in the middle of the year and stay high in Aug to Sep.

Task 4. Confirm with autocorrelation [4 marks]

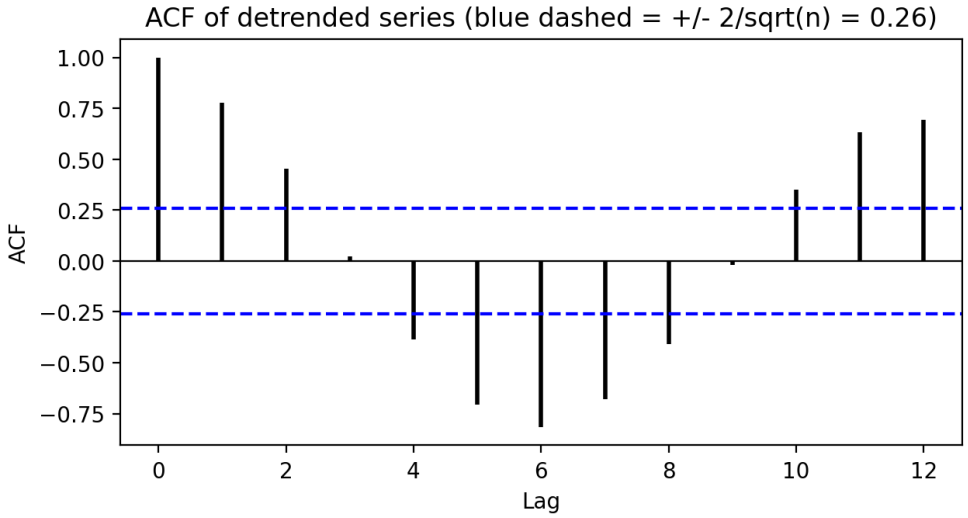
R code

```
d <- resid(lm(visits ~ 1:n))
a <- acf(d, lag.max = 12, plot = FALSE)
a$acf[13]
a$acf[7]
2 / sqrt(n)

plot(acf(d, lag.max = 12, plot = FALSE), main = "ACF of detrended series")
```

Output

ACF at lag 12	ACF at lag 6	Bound 2/sqrt(n)
0.69	-0.82	0.26



Answer:
Autocorrelation at lag 12 = 0.69, at lag 6 = -0.82, and the significance bound 2/sqrt(60) = 0.26.
Both values are far outside the bound. The lag 12 value is strongly positive, so a month looks like the same month one year earlier. The lag 6 value is strongly negative, so a month is opposite to the month 6 months away (high season vs low season). Together they confirm a strong seasonal pattern with a cycle of 12 months.

Task 5. Additive or multiplicative? [6 marks]

R code

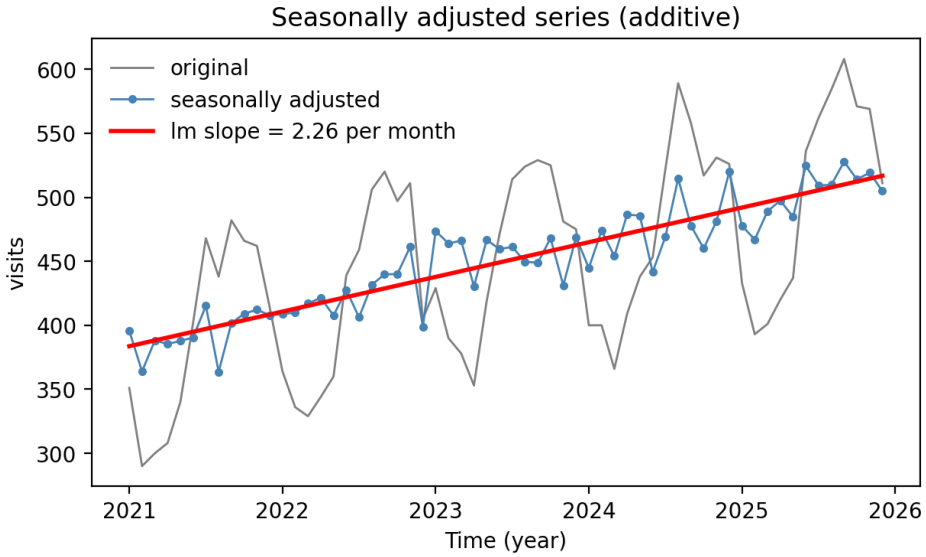
```
rng1 <- diff(range(visits[1:12]))
rng5 <- diff(range(visits[49:60]))
rng1; rng5; rng5 / rng1

dec_a <- decompose(x, type = "additive")
adjusted <- x - dec_a$seasonal
fit2 <- lm(adjusted ~ 1:n)
coef(fit2)[2]

plot(adjusted, type = "o", pch = 16, col = "steelblue",
     main = "Seasonally adjusted series", ylab = "visits")
lines(time(x), fitted(fit2), col = "red", lwd = 2)
```

Output

Range of first cycle	Range of last cycle	Ratio (last / first)	Slope after adjusting
192	215	1.12	2.26



Answer:
Range of first cycle = 482 - 290 = 192. Range of last cycle = 608 - 393 = 215. Ratio last / first = 1.12.
Type: additive. The ratio is close to 1, so the size of the seasonal swing stays about the same even though the level of visits goes up over the years. (If it were multiplicative, the swing would grow along with the level.)
After removing the seasonal part (x minus seasonal), the slope of lm(adjusted ~ 1:n) = 2.26 visits per month.
Advice to management: Plan more staff and supplies for August and September (about 80 visits above the trend) and fewer in February to April, and also plan for the steady growth of about 2.3 more visits every month.